

# Superscalar Architecture

Muhammad Tahir

Lecture 14-15

Electrical Engineering Department  
University of Engineering and Technology Lahore

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# Multiple-Issue Processor

- The processor performance

$$\text{Execution Time} = N \times CPI \times TPC$$

$N \sim$  number of instructions

$CPI \sim$  cycles per instructions

$TPC \sim$  time per cycle

- Pipelining provides
  - Instruction-level-parallelism (ILP) with scalar processor
  - $CPI \approx 1$  (practically) OR  $= 1$  (ideally) with  $TPC \downarrow$
- Further performance improvement requires  $CPI < 1$
- $CPI < 1$  requires multiple-issue architecture

# Multiple-Issue Processor Cont'd

Different types of multiple-issue processors

- **Superscalar** processor
  - Statically scheduled superscalar processor implements in-order execution
  - Dynamically scheduled superscalar processor implements out-of-order execution
- **VLIW** (Very Long Instruction Word) processor

## Program Execution Order

- **Program Order:** The order created by the compiler
- **Fetch Order:** The order of instruction fetching for execution, can be different from Program Order due to prefetching and trace-execution
- **Execution Order:** The order in which instructions are executed, can be in-order or out of order
- **Commit Order:** The order of commitment or retiring of instructions (order of updating registers and/or memory)

Prefetching = Instructions fetched before needed for execution

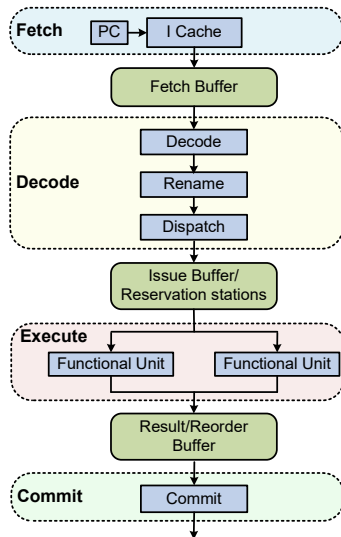
Trace execution = fetching and executing instructions along a predicted path

## Instruction Execution Order

- Orders associated with instruction execution
  - Order in which instructions are **Issued**
  - Order in which instructions are **Completed**
- Useful execution ordering combinations
  - In-order Issue with In-order Completion
  - In-order Issue with Out-of-order Completion
  - Out-of-order Issue with Out-of-order Completion

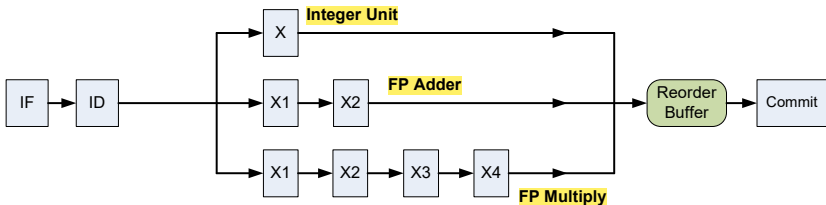
# Instruction Execution Phases

- Four phases of instruction execution
- Any two phases have an intermediate buffer
- Each buffer is critical for multiple-issue architecture
- Instruction **fetch**, **decode**, **rename** and **dispatch** are in-order
- **Execute** and **commit** can be in-order or out-of-order



## In-order Issue

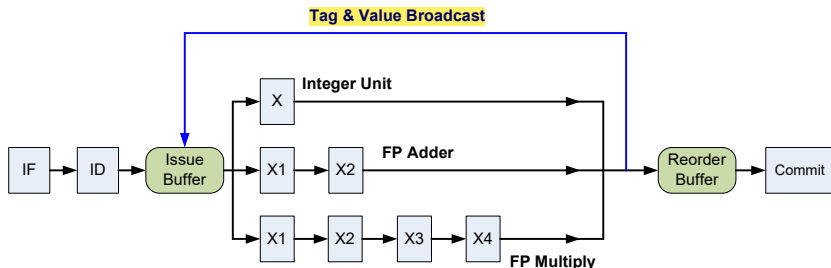
- In-order issue and out-of-order completion
- Also called static scheduling
- Use reordering buffer for in-order commit





## Out-of-Order Issue

- Out-of-order issue and out-of-order completion
- Also called dynamic scheduling
- Use issue buffer for out-of-order issue
- Use reordering buffer for in-order commit

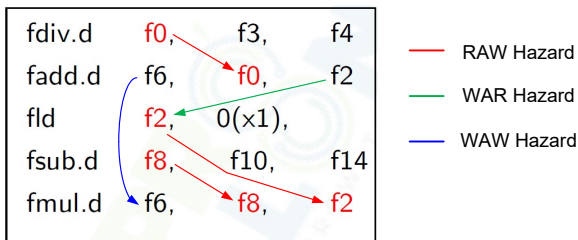


## Problem with Out-of-Order Issue

- The WAW and WAR hazards
- False data dependencies

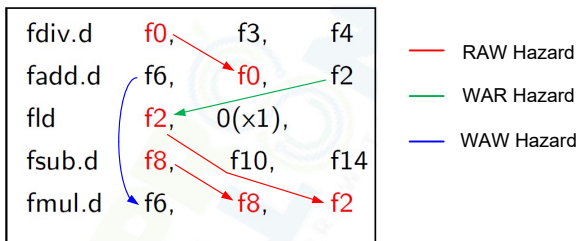
## Problem with Out-of-Order Issue

- The WAW and WAR hazards
- False data dependencies
- Example program with floating point instructions



## Problem with Out-of-Order Issue

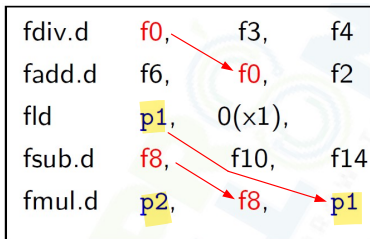
- The WAW and WAR hazards
- False data dependencies
- Example program with floating point instructions



- Assume **fdiv** takes 5 cycles, **fmul** takes 3 cycles, while other instructions take one cycle

# Register Renaming

- The solution for WAW and WAR hazards
- Use register renaming



— RAW Hazard

# In-order Execution

Is this scalar or super scalar?

- In-order issue
- No forwarding

|        | Cycles     | 1   | 2  | 3 | 4 | 5              | 6              | 7              | 8              | 9              | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
|--------|------------|-----|----|---|---|----------------|----------------|----------------|----------------|----------------|----|----|----|----|----|----|----|
| fdiv.d | <b>f0,</b> | f3, | f4 | F | D | X <sub>1</sub> | X <sub>2</sub> | X <sub>3</sub> | X <sub>4</sub> | X <sub>5</sub> | W  |    |    |    |    |    |    |

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- No forwarding

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| fdiv.d | f0, | f3, | f4 | F | D | X <sub>1</sub> | X <sub>2</sub> | X <sub>3</sub> | X <sub>4</sub> | X <sub>5</sub> | W |                |    |    |    |    |    |    |    |
| fadd.d | f6, | f0, | f2 |   | F | D              | -              | -              | -              | -              | - | X <sub>1</sub> | W  |    |    |    |    |    |    |

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| fadd.d | f6, f0, f2 |   | F | D              | -              | -              | -              | -              | - | X <sub>1</sub> | W              |    |    |    |    |    |    |
| fld    | f2, 0(x1), |   |   | F              | -              | -              | -              | -              | - | D              | X <sub>1</sub> | W  |    |    |    |    |    |

WAR



# In-order Execution

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| fdiv.d | f0, f3, f4   | F      | D | X <sub>1</sub> | X <sub>2</sub> | X <sub>3</sub> | X <sub>4</sub> | X <sub>5</sub> | W |   |                |                |                |    |    |    |    |    |
| fadd.d | f6, f0, f2   |        | F | D              | -              | -              | -              | -              | - | - | X <sub>1</sub> | W              |                |    |    |    |    |    |
| fld    | f2, 0(x1),   |        |   | F              | -              | -              | -              | -              | - | - | D              | X <sub>1</sub> | W              |    |    |    |    |    |
| fsub.d | f8, f10, f14 |        |   |                |                |                |                |                |   |   | F              | D              | X <sub>1</sub> | W  |    |    |    |    |

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| fadd.d | f6, | f0,    | f2  |        |   | F | D              | -              | -              | -              | -              | - | X <sub>1</sub> | W              |                |    |                |                |                |    |
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| fmul.d | f6, | f8,    | f2  |        |   |   |                |                |                |                |                |   |                | F              | D              | -  | X <sub>1</sub> | X <sub>2</sub> | X <sub>3</sub> | W  |

# Out-of-order Execution

scalar or super scalar?

- Out-of-order issue
- With forwarding and imprecise exceptions

|        | Cycles     | 1 | 2 | 3              | 4              | 5              | 6              | 7              | 8 | 9 | 10 | 11 | 12 | 13 |
|--------|------------|---|---|----------------|----------------|----------------|----------------|----------------|---|---|----|----|----|----|
| fdiv.d | f0, f3, f4 | F | D | X <sub>1</sub> | X <sub>2</sub> | X <sub>3</sub> | X <sub>4</sub> | X <sub>5</sub> | W |   |    |    |    |    |

imprecise exceptions = in case of an exception system can't roll back because commits were made out of order

## Out-of-order Execution

- Out-of-order issue
- With forwarding and imprecise exceptions

| Cycles |             |             |    | 1 | 2 | 3              | 4              | 5              | 6              | 7              | 8              | 9 | 10 | 11 | 12 | 13 |
|--------|-------------|-------------|----|---|---|----------------|----------------|----------------|----------------|----------------|----------------|---|----|----|----|----|
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| fadd.d | f6,         | <b>f0</b> , | f2 |   | F | D              | -              | -              | -              | -              | X <sub>1</sub> | W |    |    |    |    |

## Out-of-order Execution

- Out-of-order issue
- With forwarding and imprecise exceptions

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| fadd.d | f6, | f0,    | f2 |   | F | D              | -              | -              | -              | -              | X <sub>1</sub> | W              |    |    |    |    |
| fld    | f2, | 0(x1), |    |   |   | F              | D              | -              | -              | -              | -              | X <sub>1</sub> | W  |    |    |    |

# Out-of-order Execution

- Out-of-order issue
- With forwarding and imprecise exceptions

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| fadd.d | f6, | f0,    | f2  |        |   | F | D              | -              | -              | -              | -              | X <sub>1</sub> | W              |    |    |    |    |
| fld    | f2, | 0(x1), |     |        |   |   | F              | D              | -              | -              | -              | -              | X <sub>1</sub> | W  |    |    |    |
| fsub.d | f8, | f10,   | f14 |        |   |   |                | F              | D              | X <sub>1</sub> | W              |                |                |    |    |    |    |

# Out-of-order Execution

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| fdiv.d | f0,    | f3,    | f4  |  | F | D | X <sub>1</sub> | X <sub>2</sub> | X <sub>3</sub> | X <sub>4</sub> | X <sub>5</sub> | W              |                |                |                |                |    |
| fadd.d | f6,    | f0,    | f2  |  |   | F | D              | -              | -              | -              | -              | X <sub>1</sub> | W              |                |                |                |    |
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# Out-of-order Execution with Register Renaming

- Out-of-order issue with register renaming
- With forwarding, precise exceptions and multiple commits

|        | Cycles     | 1 | 2 | 3              | 4              | 5              | 6              | 7              | 8              | 9 | 10 | 11 |
|--------|------------|---|---|----------------|----------------|----------------|----------------|----------------|----------------|---|----|----|
| fdiv.d | f0, f3, f4 | F | D | X <sub>1</sub> | X <sub>2</sub> | X <sub>3</sub> | X <sub>4</sub> | X <sub>5</sub> | R              | W |    |    |
| fadd.d | f6, f0, f2 |   | F | D              | -              | -              | -              | -              | X <sub>1</sub> | R | W  |    |
| fld    | p1, 0(x1), |   |   | F              | D              | X <sub>1</sub> | R              | -              | -              | - | W  |    |

Is it scalar or superscalar?

Contains reorder buffer



# Out-of-order Execution with Register Renaming

- Out-of-order issue with register renaming
- With forwarding, precise exceptions and multiple commits

|        |     |        |     | Cycles | 1 | 2 | 3              | 4              | 5              | 6              | 7              | 8              | 9              | 10 | 11 |
|--------|-----|--------|-----|--------|---|---|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----|----|
| fdiv.d | f0, | f3,    | f4  |        | F | D | X <sub>1</sub> | X <sub>2</sub> | X <sub>3</sub> | X <sub>4</sub> | X <sub>5</sub> | R              | W              |    |    |
| fadd.d | f6, | f0,    | f2  |        |   | F | D              | -              | -              | -              | -              | X <sub>1</sub> | R              | W  |    |
| fld    | p1, | 0(x1), |     |        |   |   | F              | D              | X <sub>1</sub> | R              | -              | -              | -              | W  |    |
| fsub.d | f8, | f10,   | f14 |        |   |   |                | F              | D              | X <sub>1</sub> | R              | -              | -              | W  |    |
| fmul.d | p2, | f8,    | p1  |        |   |   |                |                | F              | D              | X <sub>1</sub> | X <sub>2</sub> | X <sub>3</sub> | R  | W  |

# Dynamic Scheduling Techniques

- Two key approaches for Dynamic Scheduling
  - Scoring
  - Tomasulo's Algorithm

# Tomasulo's Algorithm: Key Features

- Reservation Stations

- A **Buffer** for a functional unit to hold instructions and their **operands**
- An **operand** can hold an actual *value* or the *tag* of a reservation station entry. It can also hold a load buffer entry that will provide the value

- Common Data Bus

- Connects *functional units* and *load buffer* to reservations stations, registers and store buffer
- Responsible to transfer data *simultaneously* to *all* the stations/registers that require it

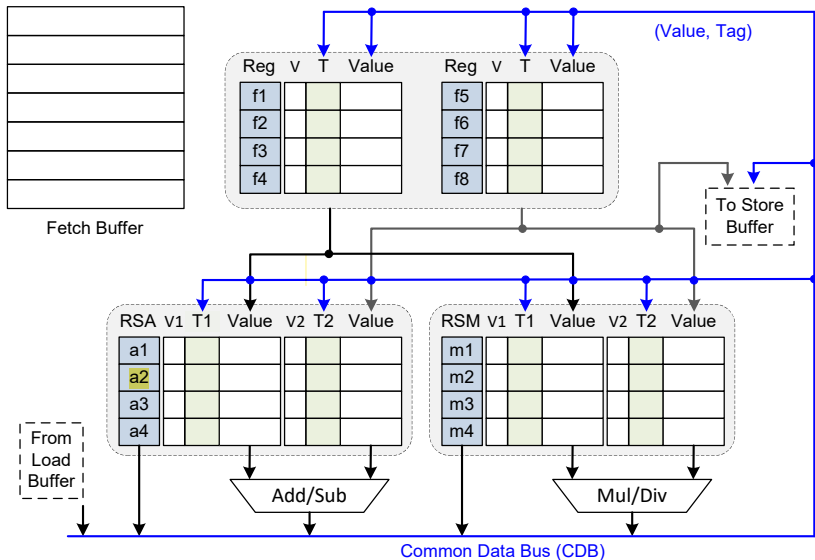
## Tomasulo's Algorithm: Key Features Cont'd

- Addressing Hazards
  - Uses (hardware) register renaming to eliminate WAR and WAW hazards
  - Operands refer to reservation stations/load buffer (data producer) for the values
  - Usually have more reservation stations than registers (for effective register renaming)

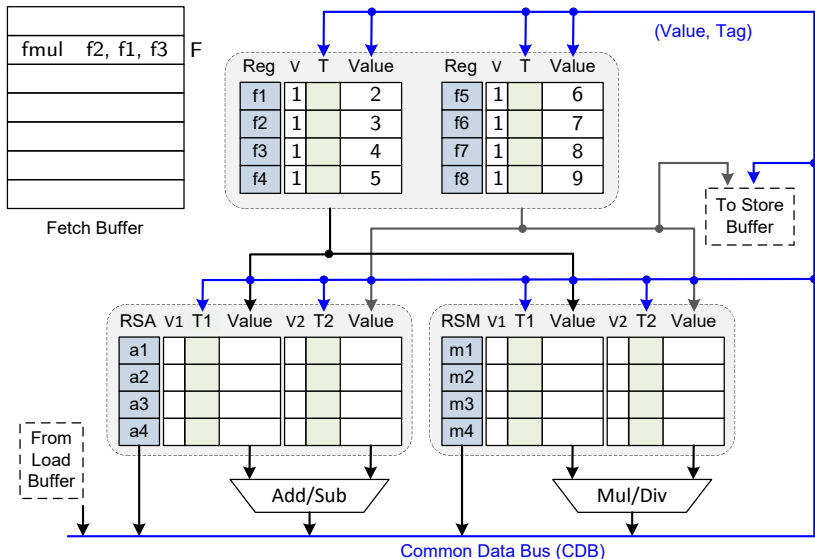
# Tomasulo's Algorithm: Pipeline Stages

- **Fetch:** Get the instructions to **fetch buffer**
- **Dispatch:** Decode instruction, perform register renaming and allocate **reservation station (RS)** (or **issue buffer**), stall in case of structural hazard (no free space in RS)
- **Execute:** Issue the instruction in RS to **functional unit (FU)**, in case of RAW hazard wait and monitor **common data bus (CDB)** for the value
- **Write:** Write the result to register (store buffer), free the RS entry

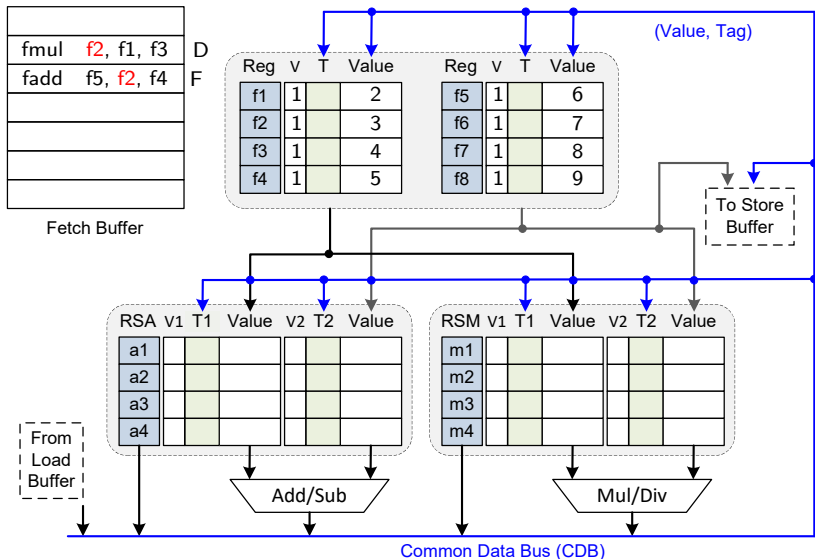
# Tomasulo's Algorithm: Block Diagram



# Out-of-Order Implementation: Cycle 1

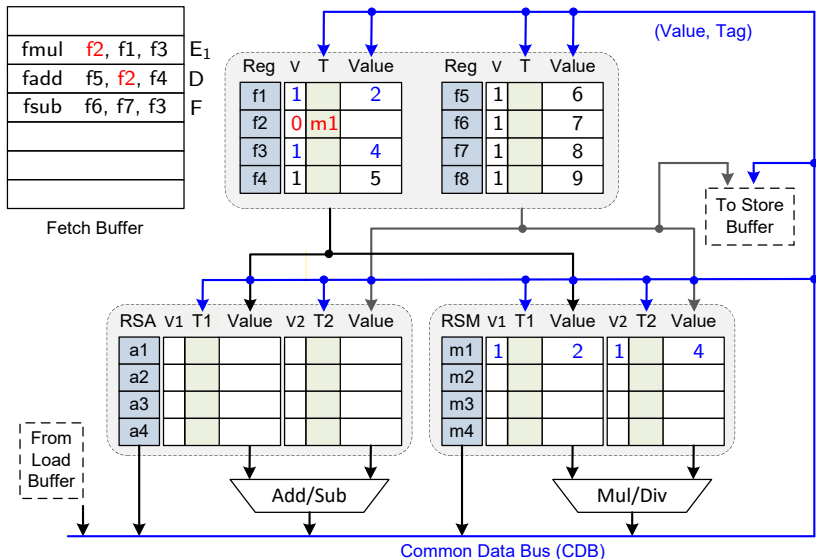


# Out-of-Order Implementation: Cycle 2

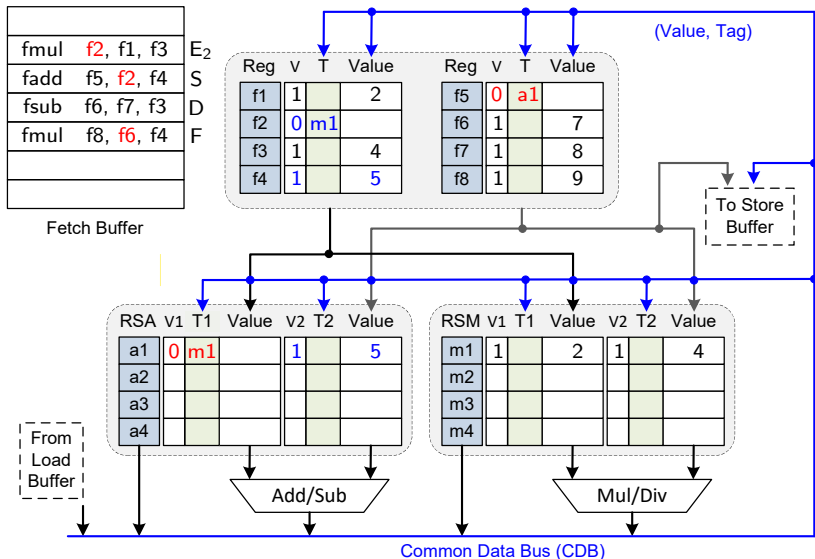




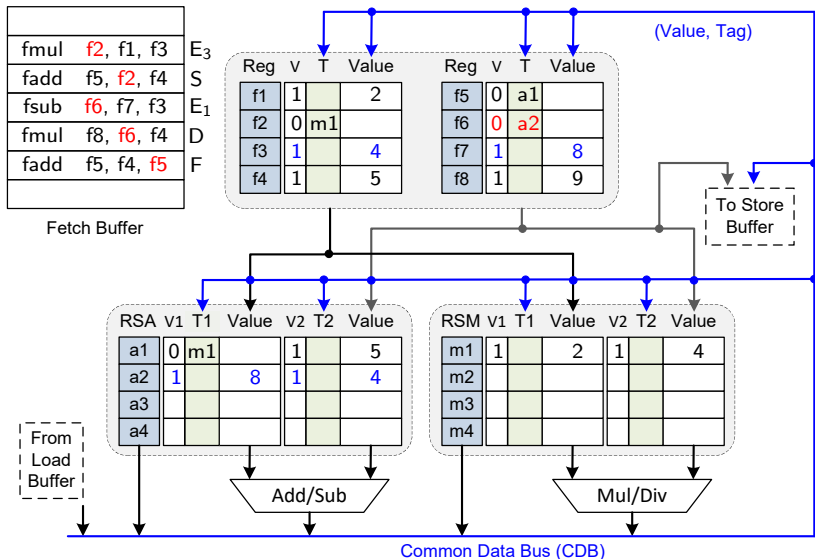
# Out-of-Order Implementation: Cycle 3



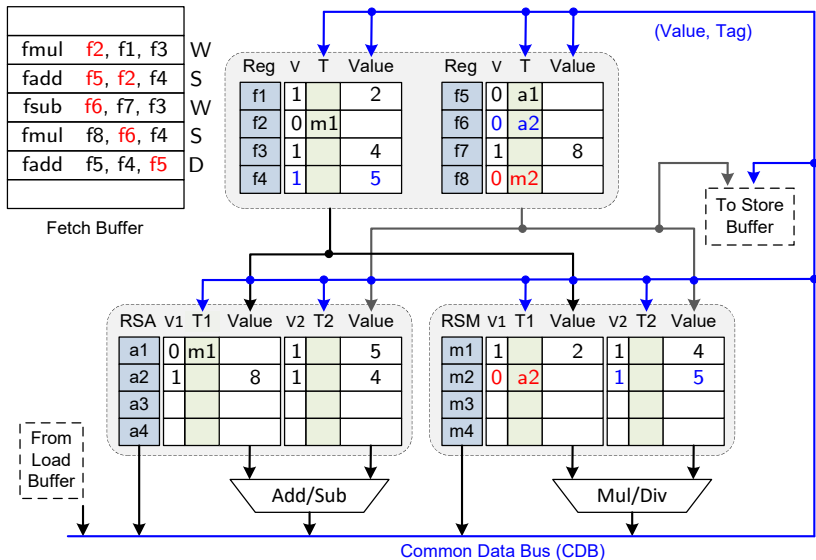
# Out-of-Order Implementation: Cycle 4



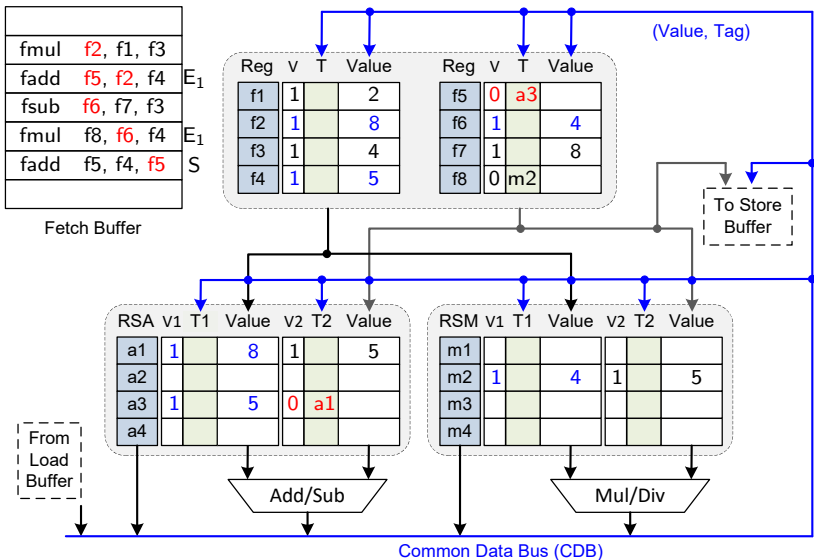
## Out-of-Order Implementation: Cycle 5



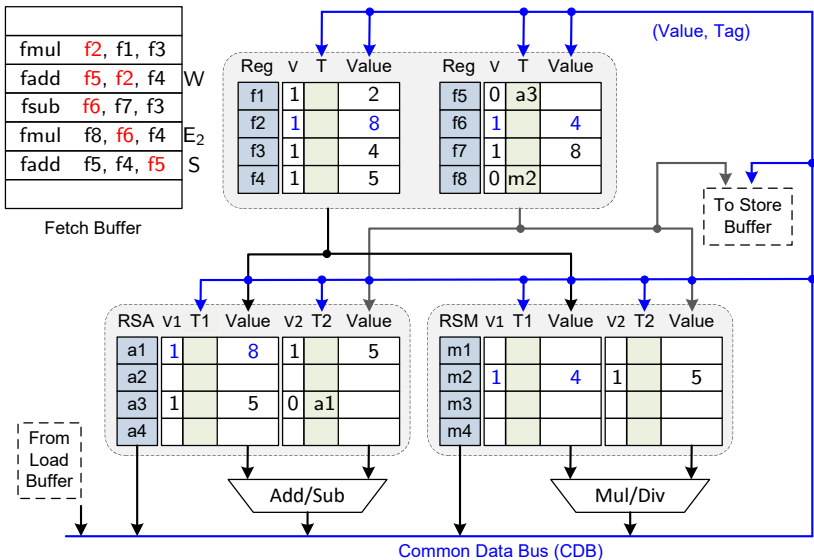
# Out-of-Order Implementation: Cycle 6



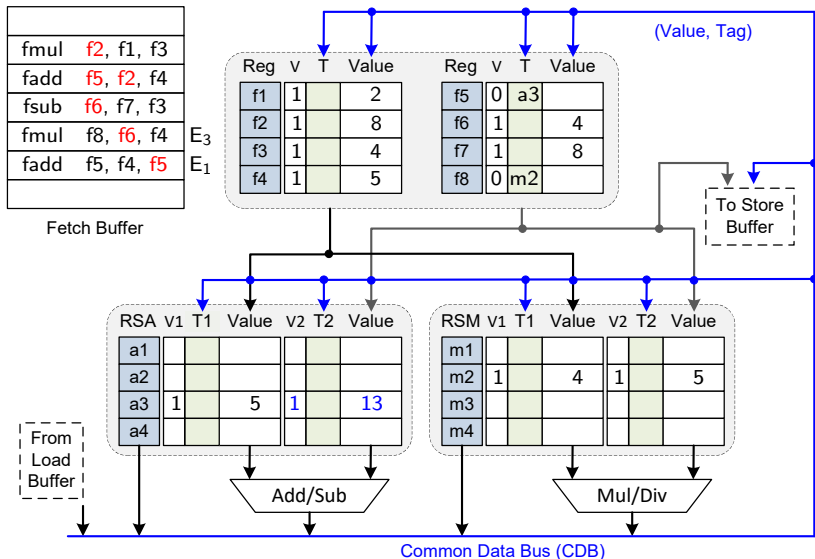
# Out-of-Order Implementation: Cycle 7



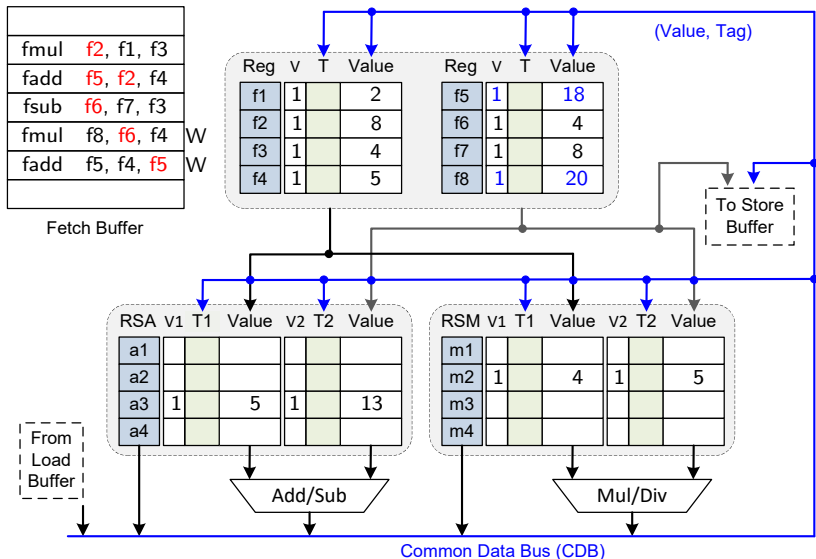
# Out-of-Order Implementation: Cycle 8



# Out-of-Order Implementation: Cycle 9



# Out-of-Order Implementation: Cycle 10





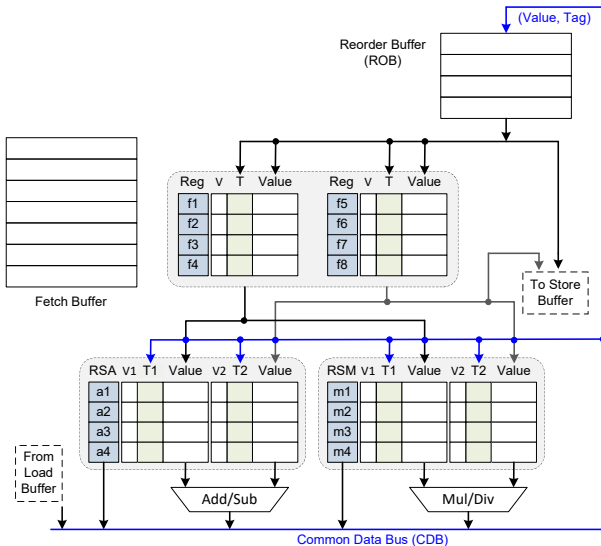
## Reorder Buffer

Using reorder buffer (ROB) allows:

- In-order commit to maintain program order state
- Precise exception handling
- Speculative execution (branch prediction is implied)



# Tomasulo's Algorithm with ROB



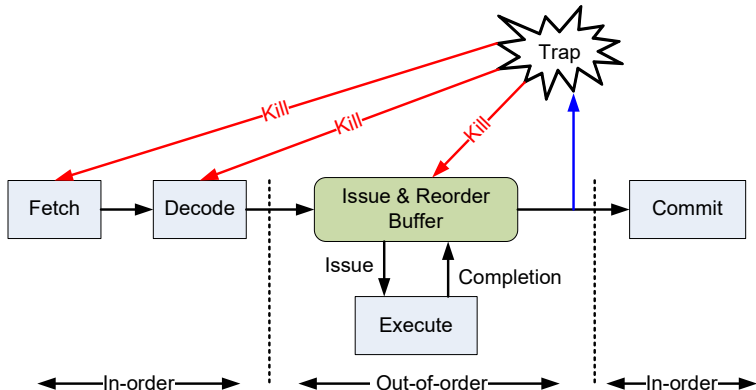
## Reorder Buffer & Speculation

- Branch is issued with prediction marking in ROB
- Speculatively fetch, decode and issue instructions based on prediction
- Branch must **resolve** before leaving ROB
  - **Resolved** as predicted: Commit the following instructions after branch
  - **Not Resolved** as predicted: Mark the following instructions in ROB as invalid

## Handling Exceptions and Interrupts

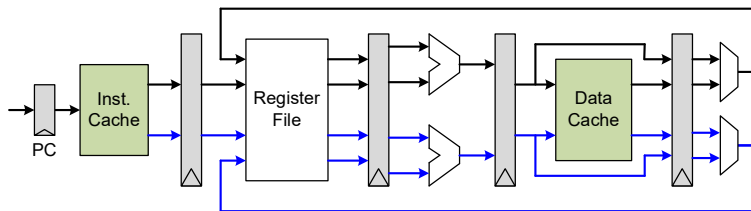
- The exceptions and interrupts are handled at commit stage
- Out-of-order commit leads to **imprecise traps**
- In-order commit leads to **precise traps**
- In-order commit requires **Reorder Buffer** or **Queue**

# Handling Exceptions and Interrupts Cont'd

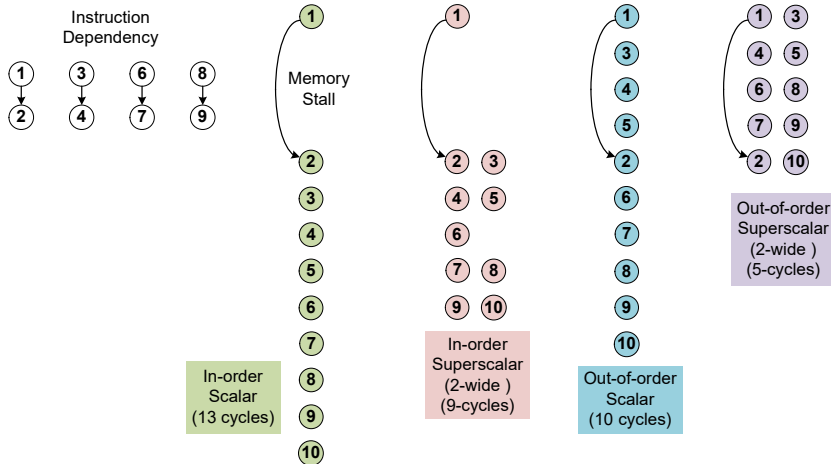


# The Superscalar: Implementation

- In-order, 2-wide pipeline



# The Superscalar IS NOT OOO





# VLIW Processor

- **VLIW processor:** **Statically** scheduled by the compiler with multiple operations packed into *very large instruction word* (VLIW)
- **VLIW compiler:**
  - Schedule operations to maximize parallelism (loop unrolling, trace scheduling etc.)
  - Avoid data hazards
- **Example VLIW processors:** **TMS320C6x DSPs** (embedded DSP with partial success), Intel Itanium (discontinued)
- **Performance limitation:** Limitations of **static scheduling**

## Suggested Reading

- Read relevant sections of Chapter 3 of [\[Patterson and Hennessy, 2019\]](#).

# Acknowledgment

- Preparation of this material was partly supported by Lampro Mellon Pakistan.

# References



Patterson, D. and Hennessy, J. (6th Edition, 2019).  
*Computer Architecture: A Quantitative Approach*.  
Morgan Kaufmann.